Name: Jojo Greear, Ethan Williamson, Hayden Cornelius, and Callysta Riggans

Class: 2nd Period HS Engineering

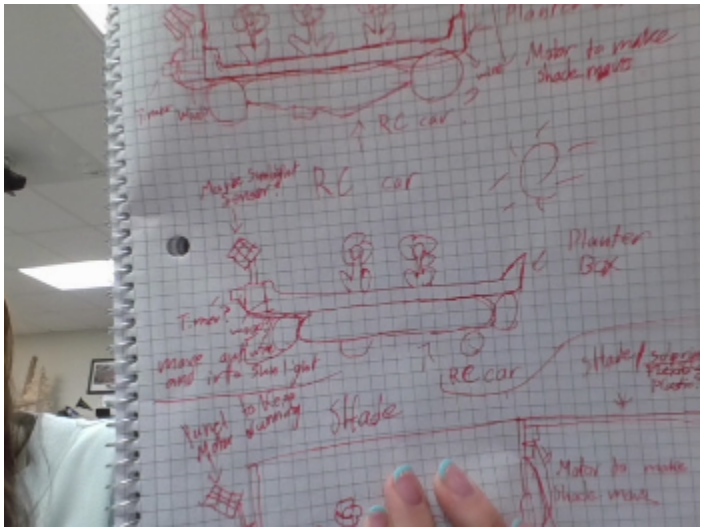
MyDesign Portfolio

Element D Initial Template

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

RC Car



Date: December 2 2024
Description: Device could be on a timer or use heat motion to go where the most sunlight is, while on an RC car.

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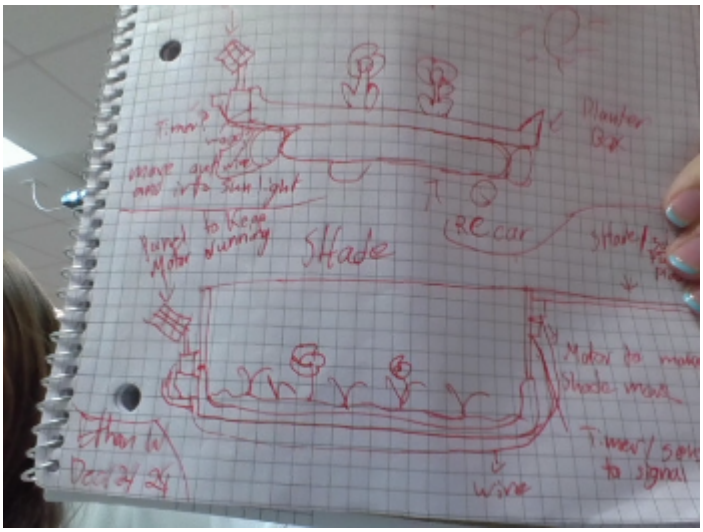
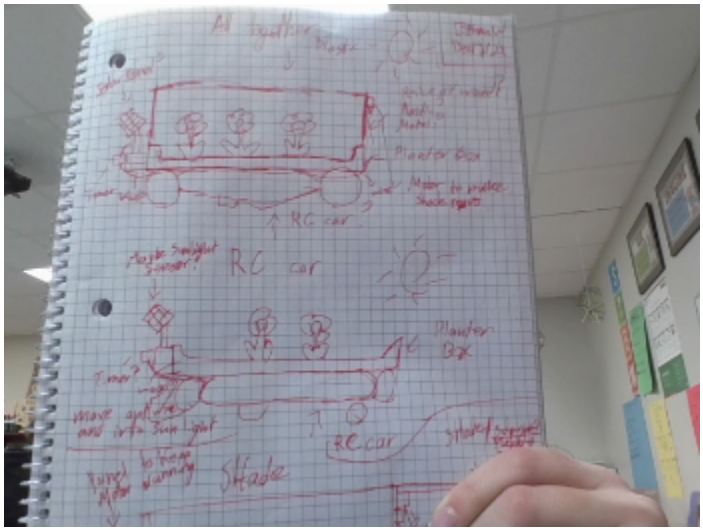
Shade



Date: December 2 2024
Description: Use 3d printer plastic, flexible plastic to use for shade; we move the shade off of and above the plant based on the time.

First two combined



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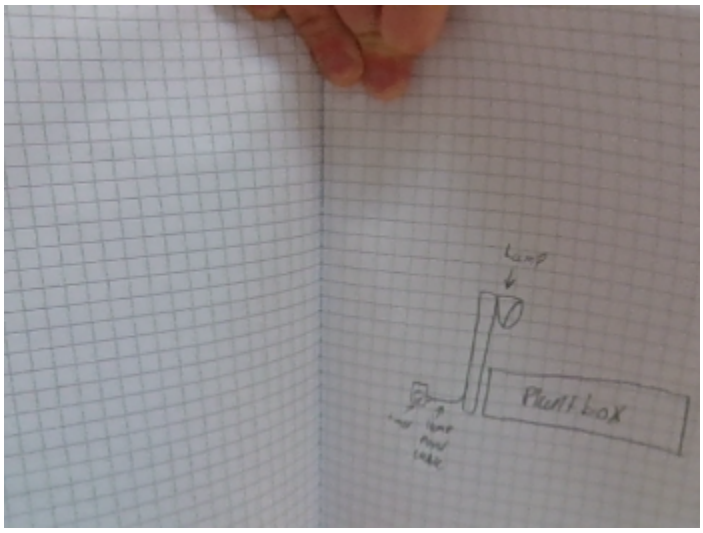
Date: December 2 2024
Description: Use a RC car and build a shade to move into and out of the sunlight.

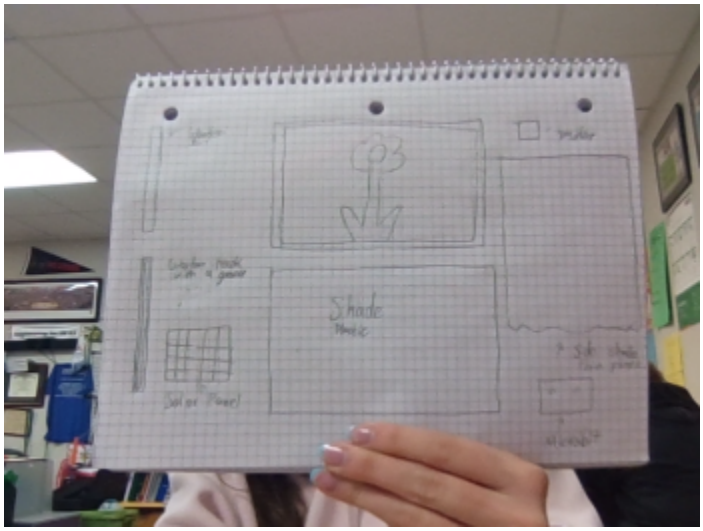
Indoor



Date: December 2 2024

Description: Use a timer connected to a light source to activate and deactivate light at a certain time of the day.

Design Idea Comparison and Selection *(D1, D2)*



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We decided to choose the second design with some modifications because it will be the most efficient. We think it will be more efficient because it is less complicated than having a lot of different aspects like our 3rd idea and it can be programmed to work in less sunny places unlike the 1st design.

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Design Blueprint *(D4)*

	weight	Option 1	Option 2	Option 3	Option 4
Device functionality	25%	3	5	2.5	3
Required size	15%	3	4	2	2
Easy to use	25%	4	4	3	4
Works in different environment	15%	2	4	3	1
Specific to each plant	10%	4	5	5	4
Cost	10%	4	2.5	1	4
score		3.3	4	2.75	3

Supply List and Costs:

Item	**Cost each**	**# needed**	**Total**
Solar panel	$10-$50	1	$10-$50
Piece of shaded plastic	free	1	free
poles	free	4	free
Shaded tarp	free	4	free
Motor	Free/$60	1	free/60$
Plant platform	free	1	free
Track with a groove	free	3	free

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Timer	Free	1	Free
Total Cost			$100-$120

Plan of Action *(D3)*

First...

We will build our base structure/foundation, using wood.

Second...

We will install a 20x20 plant tray onto our platform.

Third...

We will attach our shade to the top of our device.

Fourth...

We will add our motors and wires that will move our platform back and forth.

Fifth...

We will program the device to move at specific times of the day.

Finally we will test and make modifications to our device.